

Analyzing Thematic Alignment in Drug Safety

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1 Introduction

Scientific journals define their editorial focus through an *Aims & Scope* statement, which describes the topics and research areas they intend to cover. However, the articles they publish do not always perfectly reflect this stated scope, and the journal’s focus may also change over time. Sentence embeddings provide a practical way to examine this relationship, as they allow us to capture and study the semantic meaning of texts and make it possible to compare large collections of scientific publications by measuring their similarity [3].

In this project, this approach is applied to *Drug Safety*, a well-established journal whose abstracts are accessible through PubMed.

The study covers articles published between 2015 and 2025 and measures their similarity with the journal’s *Aims & Scope* using SentenceTransformer embeddings and cosine similarity, following the Sentence-BERT approach [3]. The analysis then explores how this alignment changes over time and identifies both the articles that differ most from the journal’s stated focus and those that are most closely aligned with it. The project also includes a comparison with a naive BERT baseline to evaluate the choice of embedding model and, finally, BERTopic is used to identify the main thematic clusters in the corpus and examine how their prevalence changes over time.

2 Research Question and Methodology

The analysis is guided by the following research question:

To what extent are the articles published in Drug Safety thematically aligned with the journal’s stated Aims & Scope, and how has this alignment evolved between 2015 and 2025?

2.1 Data Collection

Metadata and abstracts from *Drug Safety* were retrieved through the PubMed NCBI E-utilities API, covering the period from 2015 to 2025.

Among the 1,355 records initially collected, 1,129 contained a valid title, publication year, and a non-empty abstract of sufficient length and were therefore retained for the analysis. A small number of records were dated 2026 because PubMed had indexed them as “online first” before their formal publication, reflecting the normal indexing process rather than an error in the data. The resulting incomplete coverage of 2026 was examined and managed during the exploratory analysis described below.

2.2 Exploratory Data Analysis

Before computing embeddings and alignment scores, an exploratory analysis of the raw corpus was conducted to examine the data and identify potential issues for the subsequent analysis.

Abstract length was examined first. Abstracts contained an average of 1,839 characters, with a median of 1,816 and a maximum of 4,006. Since the embedding model operates on tokens rather than characters, length was also measured using the model’s tokenizer. Overall, 91.05% of the abstracts exceeded the 256-token input limit of `all-MiniLM-L6-v2`, with an average length of 397.8 tokens and a maximum of 1,000. Direct truncation would therefore have removed part of most abstracts. To preserve this information, long abstracts were divided into sentence-based chunks, whose embeddings were later combined using a token-weighted average, as described in Section 2.4.

Publication years were also checked. Although data collection targeted the period from 2015 to 2025, 39 records were dated 2026 because they had been indexed by PubMed as “online first”. This number was considerably lower than the annual output observed between 2015 and 2025, which ranged from 89

to 115 articles per year, indicating that 2026 represented only a partial publication year at the time of data collection. The 39 records were therefore excluded from the analysis.

Finally, duplicate titles were examined. One repeated title was found, but comparison of the corresponding abstracts showed that the two records were not duplicates: both were editorial correction notices referring to the same original article, with one reporting a figure correction and the other referring to the corrected DOI. Correction notices were then examined more broadly because an initial run of the alignment pipeline showed that several of the lowest-scoring records had titles beginning with “Correction to:”. These short editorial texts differ substantially from research abstracts in both content and purpose, making their low similarity scores difficult to interpret as evidence of thematic misalignment. All 16 correction notices identified in the corpus, including the two records sharing the same title, were therefore excluded from the final dataset.

After these exclusions, the final corpus used for all subsequent analyses consisted of 1,074 articles.

2.3 Text Preprocessing

The journal’s *Aims & Scope* statement was manually collected from the official publisher’s webpage [5], as this information is not available through a structured API. Before computing the embeddings, the same preprocessing procedure was applied to both the abstracts and the *Aims & Scope* text, with multiple spaces, tabs, and line breaks collapsed into single spaces while preserving the original wording and casing, since the embedding model is designed to process natural text without requiring extensive preprocessing.

2.4 Embedding and Chunking

In this project, semantic representations were generated using the pre-trained SentenceTransformer model `all-MiniLM-L6-v2` [3]. Sentence-BERT was preferred to raw BERT because it is specifically designed to produce embeddings suitable for semantic similarity comparisons, while `all-MiniLM-L6-v2` was selected for its balance between computational efficiency and embedding quality [4]. The suitability of Sentence-BERT for this specific corpus is further assessed empirically through a comparison with a naive BERT baseline in Section 3.4.

As discussed previously, most abstracts exceed the model’s input limit of 256 tokens, meaning that encoding them directly would result in substantial truncation and loss of information. For this reason, sentence-based chunking was adopted using NLTK’s sentence tokenizer. Consecutive complete sentences were added to each chunk until the model’s token budget was reached, so that sentence boundaries were preserved while ensuring that every chunk remained within the input limit.

Formally, let d_i denote the abstract of article i , represented as a set of sentence-based chunks $d_i = \{c_{i1}, c_{i2}, \dots, c_{im}\}$. Each chunk is encoded independently as $\mathbf{v}_{ij} = f(c_{ij})$ and normalized to unit length, after which the document-level representation is obtained through a weighted average:

$$\mathbf{v}_i = \frac{\sum_{j=1}^m w_{ij} \hat{\mathbf{v}}_{ij}}{\sum_{j=1}^m w_{ij}}, \quad \hat{\mathbf{v}}_{ij} = \frac{\mathbf{v}_{ij}}{\|\mathbf{v}_{ij}\|} \quad (1)$$

where w_{ij} corresponds to the number of tokens in chunk c_{ij} . Weighting by token count allows longer chunks to contribute proportionally more to the document representation, while normalizing each chunk embedding before averaging prevents differences in vector magnitude from influencing these weights. The same procedure was applied to the *Aims & Scope* text and reduces to a single embedding when the text fits within one chunk.

A final check confirmed that 92.3% of the abstracts in the cleaned corpus required more than one chunk, while the longest abstract was divided into five, supporting the need for a chunking strategy rather than direct truncation.

2.5 Alignment Score Computation

Given the document-level embedding $\mathbf{v}_i$ of article i and the embedding $\mathbf{v}_s$ of the journal’s *Aims & Scope*, the thematic alignment was measured using cosine similarity:

$$\text{Alignment}(i) = \cos(\mathbf{v}_i, \mathbf{v}_s) = \frac{\mathbf{v}_i^\top \mathbf{v}_s}{\|\mathbf{v}_i\| \|\mathbf{v}_s\|}. \quad (2)$$

The resulting score reflects the semantic proximity between each abstract and the journal’s stated scope, with higher values indicating greater alignment. Cosine similarity was chosen because it compares

the orientation of the embedding vectors independently of their magnitude and is widely used for semantic similarity comparisons with sentence embeddings [3].

2.6 Outlier Detection

Articles with unusually high or low alignment scores were identified using a z -score criterion, which measures how far each observation lies from the mean of the alignment score distribution:

$$\left| \frac{\text{Alignment}(i) - \mu}{\sigma} \right| > 2.0, \quad (3)$$

where μ and σ denote the mean and standard deviation of the alignment scores. Because the absolute z -score is used, the criterion captures observations at both ends of the distribution, including articles with unusually low alignment and those with unusually high alignment. A threshold of 3.0 was initially considered, but its application to this corpus identified only two articles and, since it was too restrictive for a meaningful qualitative analysis, the threshold was consequently set to 2.0, which corresponds approximately to the outer 5% of observations when the underlying distribution is normal.

Since this probabilistic interpretation depends on approximate normality, the distribution of alignment scores was examined before interpreting the resulting outliers. The Shapiro–Wilk test rejected the null hypothesis of normality ($p = 0.0346$), but formal normality tests can be sensitive to small deviations in relatively large samples [6]. Since in this case skewness (-0.13) and kurtosis (-0.24) were both close to zero and the distribution showed a unimodal and only mildly asymmetric shape, the z -score criterion was kept as a descriptive method for identifying observations in the tails of the distribution rather than as a formal test for anomalous articles.

To assess whether the identified cases depended strongly on this choice, the analysis was complemented by the interquartile range rule with $k = 1.5$, which provides a distribution-free alternative to the z -score criterion, and the agreement between the two methods was evaluated using the Jaccard similarity coefficient:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad (4)$$

where A and B represent the sets of articles identified by the two criteria.

Importantly, the articles identified through these thresholds are treated as candidates for further investigation rather than as confirmed anomalies: their alignment scores indicate that they occupy relatively extreme positions within the observed distribution, while determining whether this reflects a meaningful thematic divergence from the journal’s scope requires examining their actual content.

3 Experimental Results

3.1 Score Distribution and Outlier Counts

Alignment scores have a mean of 0.461, a median of 0.465, and a standard deviation of 0.104, with values ranging from 0.094 to 0.741. As shown in Figure 1, the distribution is unimodal and only slightly asymmetric, consistent with the near-zero skewness and kurtosis reported before.

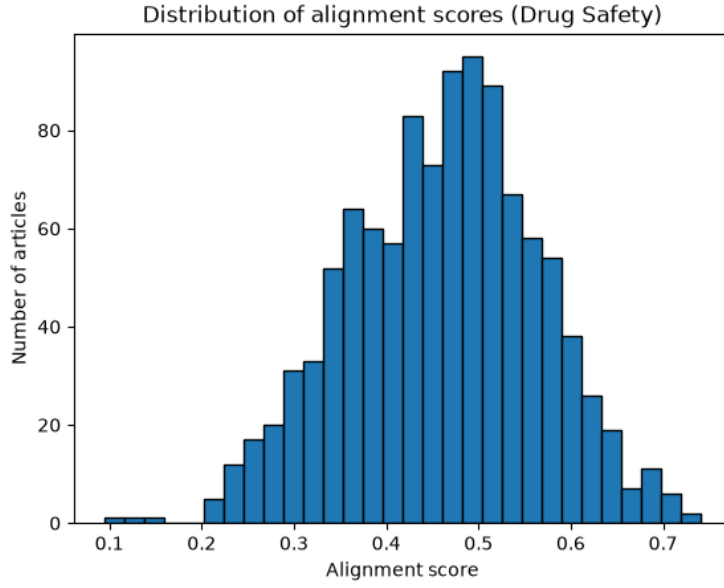


Figure 1: *Distribution of alignment scores across the 1,074-article corpus.*

Applying the z -score criterion with a threshold of 2.0 identifies 49 articles, corresponding to 4.56% of the corpus, while the IQR criterion identifies only 3, all of which are also included among the z -score cases. The two methods therefore agree on the most extreme observations, while the z -score criterion additionally captures 46 “atypical” articles. The resulting Jaccard similarity is 0.061, reflecting the difference in set size rather than disagreement between the two methods, since the three articles identified by the more conservative IQR criterion are fully contained within the larger set.

3.2 Temporal Alignment Trend

Figure 2 shows the evolution of the mean alignment score over the study period.

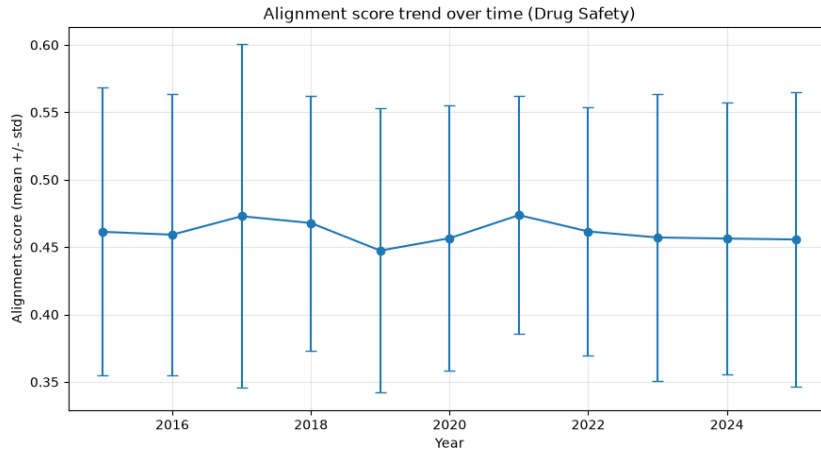


Figure 2: *Mean alignment score by publication year from 2015 to 2025. Error bars represent ± 1 standard deviation of article-level alignment scores within each year.*

The yearly mean alignment scores remain relatively stable between 2015 and 2025, with only small fluctuations and no clear increasing or decreasing trend, which suggests that, at an aggregate level, the journal maintained a broadly consistent degree of semantic alignment with its Aims & Scope throughout the period considered. However, the yearly mean does not describe the full distribution of article-level scores, and for this reason the figure also reports ± 1 standard deviation for each year. The error bars show that there is considerable variation among articles published in the same year, even when their

average alignment remains stable. They represent the dispersion of individual alignment scores around the yearly mean, rather than confidence intervals or uncertainty about the mean.

The absence of a clear trend in the yearly averages does not necessarily imply that the journal’s content remained unchanged over time, since a stable overall mean could potentially hide changes in the types of topics being published: for example, some themes could become more frequent while others decline, without substantially changing the average alignment score. This motivated the subsequent BERTopic analysis, which was used to examine the thematic structure of the corpus and its evolution over time. In this way, the topic analysis complements the aggregate alignment trend by investigating whether apparent temporal stability at the journal level masks changes in its thematic composition.

3.3 Most and Least Aligned Articles

The three highest-scoring articles, with alignment scores of 0.741, 0.727, and 0.716, focus on pharmacovigilance methods and address topics such as decision-making frameworks, meta-analysis, and reporting standards for disproportionality analysis. They closely reflect the terminology and themes described in the journal’s *Aims & Scope*, providing a coherent interpretation of their position at the upper end of the alignment distribution.

After the removal of editorial correction notices discussed in Section 2.2, the three least-aligned articles in the final corpus were *Performance and Reproducibility of Large Language Models in Named Entity Recognition* (0.094), *Suspected Adverse Effects After Human Papillomavirus Vaccination: A Temporal Relationship Between Vaccine Administration and the Appearance of Symptoms in Japan* (0.124), and *Narrative Search Engine for Case Series Assessment Supported by Artificial Intelligence Query Suggestions* (0.158). The first and third articles are centered on computational and artificial intelligence methods, which helps explain their distance from the journal’s pharmacovigilance focus, while the second is directly related to vaccine safety but is framed around a highly specific clinical context and terminology. These results suggest that a low alignment score does not necessarily indicate that an article falls outside the journal’s scope, as it may reflect either genuine thematic distance or a more specialized way of addressing an otherwise relevant topic, a distinction further explored through the topic-level analysis in Section 3.5.

3.4 Model Choice Validation

The alignment analysis presented above relies on Sentence-BERT embeddings, which were selected because they are specifically optimized for semantic similarity tasks. To assess whether this choice meaningfully affects the results on the present corpus, alignment scores were recomputed using a naive baseline based on `bert-base-uncased`, where a single representation for each text was obtained by mean-pooling its token-level embeddings [3]. The comparison was performed on a fixed random sample of 250 articles, with the random seed held constant for reproducibility. Naive BERT mean pooling produced a significantly higher average alignment score than Sentence-BERT (0.841 compared with 0.461), but lower variability (standard deviation 0.025 compared with 0.100), a pattern consistent with the anisotropy of raw BERT embeddings, which tend to occupy a relatively narrow region of the embedding space and can, therefore, produce high cosine similarities even between texts that are semantically different [1]. The two methods were only moderately correlated ($r = 0.339$, $p < 0.001$), indicating that the choice of embedding method affects the relative alignment assigned to individual articles rather than simply rescaling otherwise equivalent scores. Thus, Sentence-BERT provides greater discrimination among documents on this corpus, supporting its use for the main alignment analysis.

3.5 Topic-Level Alignment

BERTopic [2] identified 24 thematic topics together with a noise cluster containing articles that could not be assigned to a sufficiently dense group. Since a large number of topics was found, rather than reporting all clusters in the main text, the following table presents those most relevant to the qualitative inspection of the highest- and lowest-aligned articles, together with the noise cluster. The complete topic table is available in the project repository.

Topic	Count	Mean score	Diff. from overall	Representative keywords
−1 (noise)	300	0.491	+0.030	–
2	75	0.344	−0.117	vaccine, covid19, vaccines, vaccination, dose
4	48	0.409	−0.052	model, learning, machine, pharmacovigilance
8	35	0.585	+0.124	pharmacovigilance, pv, activities, countries
16	15	0.566	+0.106	disproportionality, analyses, subgroup, causal

Table 1: Selected BERTopic clusters relevant to the qualitative analysis, together with their alignment score statistics.

The noise cluster contains 300 articles (27.9% of the corpus), whose average abstract length is close to that of the articles assigned to a topic (1,881 compared with 1,850 characters), suggesting that their unassigned status is not primarily related to text length. Their mean alignment score of 0.491 is also slightly higher than the mean of 0.449 observed among the remaining clustered articles, indicating that unclustered articles are not, on average, poorly aligned with the journal’s scope. Instead, the relatively large noise cluster appears to reflect the thematic heterogeneity of the corpus, where many articles address narrow or specific subjects that do not form sufficiently dense clusters of their own.

Cross-referencing the 49 articles identified by the z-score criterion with their topic assignments provides a clearer picture of where unusually high and low alignment scores occur: of these 49 articles, 28 fall below the lower threshold and 21 above the upper threshold.

Topic 4 is mostly characterized by AI and machine learning methods, with representative terms including *model*, *learning*, *machine*, *pharmacovigilance*, and *performance*. Its mean alignment score (0.409) is below the corpus average (0.461), and two of the three least-aligned articles discussed in Section 3.3 are assigned to this topic. However, the presence of *pharmacovigilance* among its representative terms shows that the topic remains directly connected to the journal’s field, and these low scores should be interpreted as lower semantic similarity to the general wording of the journal’s *Aims & Scope*, rather than as evidence that they fall outside its thematic scope.

Topic 2 displays a similar pattern: centered on vaccine safety, it also has a mean alignment score below the corpus average, and the HPV vaccine article identified among the three least-aligned cases in Section 3.3 belongs to this topic, linking this individual low-scoring case to a wider thematic area that also shows relatively low alignment.

At the other end of the distribution, Topic 8 has one of the highest average alignment scores and is characterized by vocabulary closely related to pharmacovigilance. The highest-scoring article, *Navigating Diverging Perspectives* (0.741), belongs to this topic.

However, the three highest-scoring articles are not concentrated in a single topic: *Summarising the Evidence for Drug Safety* (0.727) is assigned to the noise cluster, and *READUS-PV* (0.716) belongs to Topic 16, which is characterized by terms such as *disproportionality*, *analyses*, and *subgroup*. High alignment therefore appears across different thematic areas, meaning that articles can closely reflect the journal’s stated scope through different aspects of pharmacovigilance.

3.6 Topic Evolution Over Time

The temporal distribution of the topics provides a different perspective on the relatively stable aggregate alignment observed across the corpus. Although the overall alignment score does not show a clear temporal trend, this does not imply that the thematic composition of the journal remained unchanged.

In fact, as shown in Figure 3, several topics vary considerably over time, indicating changes in the subjects represented in the journal during the study period. The figure displays the annual frequency of each topic, with the noise cluster excluded because it does not represent a coherent theme. Each row is normalized independently to highlight changes within topics despite differences in their overall size.

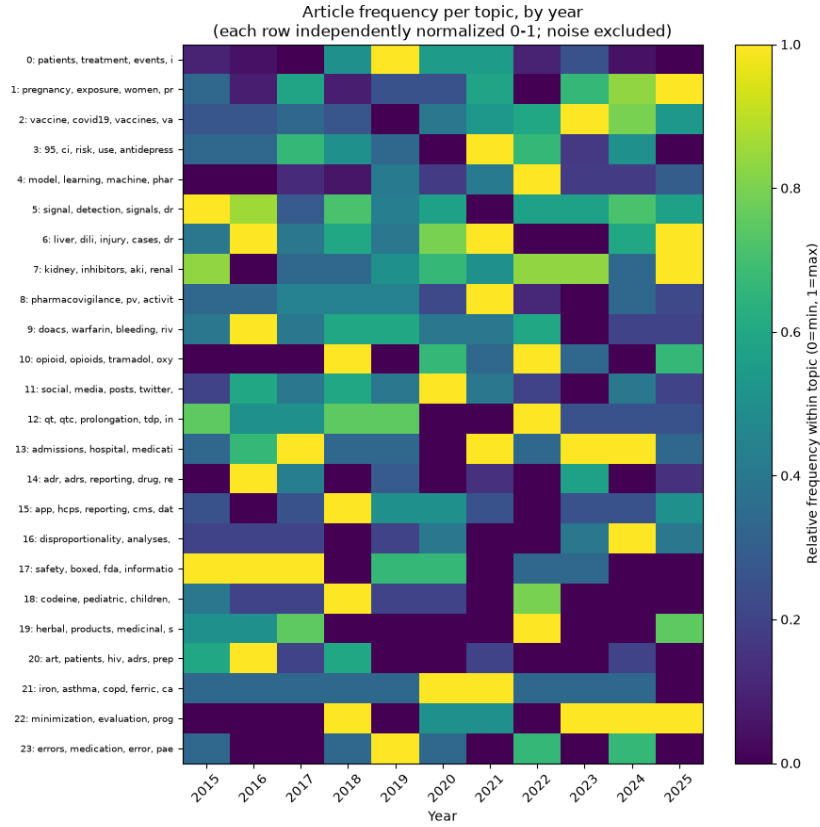


Figure 3: Article frequency per topic and year. Each row is independently normalized between its minimum and maximum frequency, while the noise cluster is excluded.

Topic 4 provides a particularly relevant example, as it is mainly associated with AI and machine learning methods and contains two of the three least-aligned articles discussed earlier. Figure 4 shows its annual number of articles: the topic is only sparsely represented in the first years of the corpus, becomes more frequent in later years, and reaches a clear peak in 2022. Its representative vocabulary also changes over time: machine-learning-related terms become prominent in 2021 and 2022, while terms such as *LLMs* and *GPT* appear among its representative keywords in 2025.

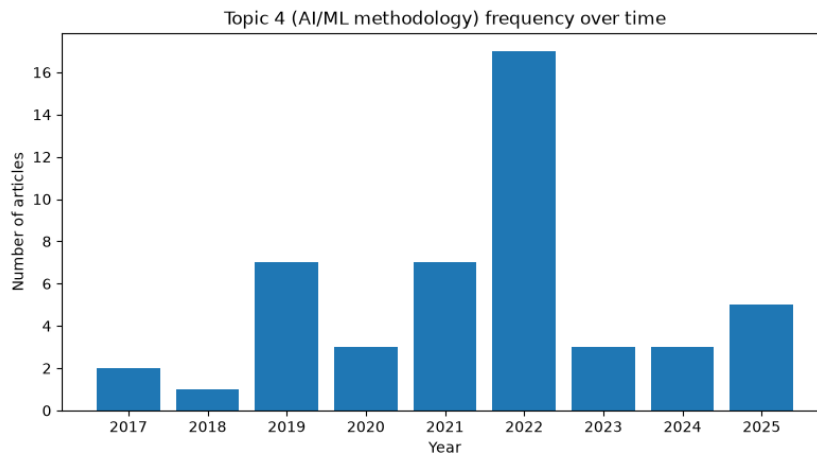


Figure 4: Article frequency, Topic 4.

Topic 2, centered on vaccine safety, provides another example of changing thematic composition, with a marked increase after 2020 and its highest frequency in 2023. Other topics also show periods of greater or lower activity, although fluctuations in smaller clusters should be interpreted cautiously because they

are based on relatively few articles.

Overall, the topic analysis shows that a relatively stable aggregate alignment can coexist with changes in the journal’s thematic composition, since the temporal alignment score and the topic-frequency analysis describe two different aspects of the corpus: the former measures how closely articles align with the journal’s stated scope, while the latter shows how the subjects represented within that scope change over time.

4 Concluding Remarks

This study examined the thematic alignment between articles published in *Drug Safety* and the journal’s stated *Aims & Scope*. Across 1,074 articles published between 2015 and 2025, the average alignment score was 0.461, with no clear systematic change over time.

However, the topic analysis demonstrated that stable overall alignment can coexist with changes in thematic composition. Topics varied in both their average alignment and their prevalence over time, with Topic 4, associated with AI and machine learning methods, providing a clear example of a thematic area that became more prominent in the later years of the corpus. Thus, the analysis showed that the journal maintained a relatively stable relationship with its stated scope while the subjects and methods represented within it evolved.

Furthermore, the comparison with naive BERT supported the use of Sentence-BERT, which provided greater differentiation among articles. At the same time, the qualitative and topic analyses showed that low similarity should not automatically be interpreted as evidence of thematic mismatch. The least-aligned articles remain connected to the journal’s field, but their specialized terminology and methodological focus may be less similar to the general wording of the *Aims & Scope*.

The main limitations of the project concern the use of a short and general *Aims & Scope* as the reference text and a pretrained embedding model without domain-specific fine-tuning. Future work could explore biomedical embedding models, richer representations of editorial scope, and comparisons across multiple journals.

AI Usage Disclaimer

For transparency, generative AI tools were used during the development of this project.

Claude (Anthropic) was used as a project structure and code assistant. All code was personally run, reviewed, and corrected by the author before use.

ChatGPT (OpenAI) was used to help draft the written report and to assist with LaTeX formatting issues, such as tables and other formatting elements.

The author takes full responsibility for the content, methodological choices, and conclusions of this work.

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